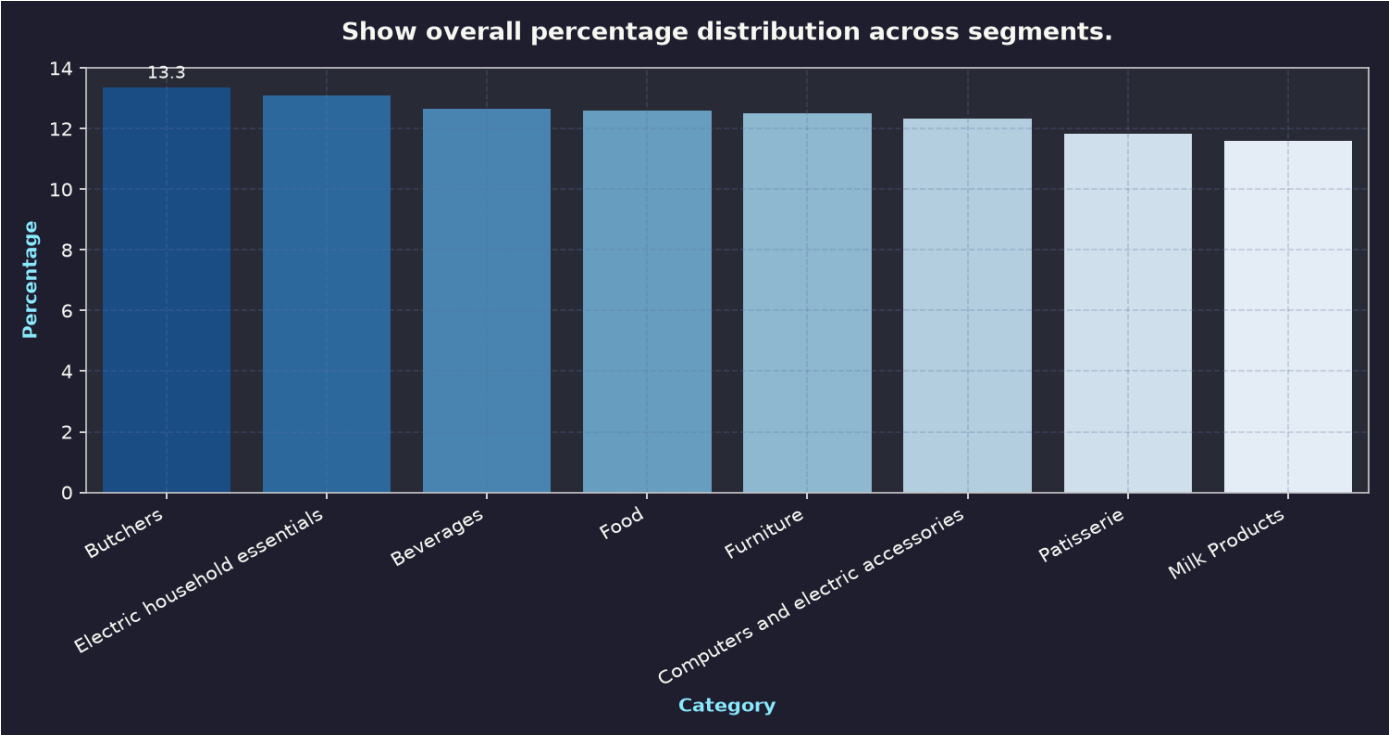


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** "Show overall percentage distribution across segments."
- **Answer:** The total spend is distributed across the eight product **segments (Category)** as follows:

Segment	% of Total Spend
Butchers	13.35 %
Electric household essentials	13.10 %
Beverages	12.66 %
Food	12.59 %
Furniture	12.52 %
Computers and electric accessories	12.33 %
Patisserie	11.84 %

| Milk Products | **11.61 %** |

These percentages sum to $\approx$ **100 %**, indicating each segment's share of overall spending.

2. Dataset Overview & Data Quality Profile

| Attribute | Detail |

|-----|-----|

| **Rows / Columns** | 12,575 × 11 |

| **Duplicate rows** | 0 (none) |

| **Missing values** | • **Item** – filled with “Unknown”
• **Price Per Unit**, **Quantity**, **Total Spent** – filled with column medians
• **Discount Applied** – filled with **False** |

| **Potential date column** | **Transaction Date** (string → datetime) |

| **Key categorical column** | **Category** (segment) |

| **Key numeric column** | **Total Spent** (used for distribution) |

3. Data Cleaning & Validation Summary

| Step | Action | Rationale |

|-----|-----|-----|

| **Copy** | `result = df.copy()` | Preserve original data |

| **Deduplication** | `drop_duplicates()` | Ensure each transaction is unique |

| **Whitespace trim** | Strip all object columns | Prevent mismatched string values |

| **Date conversion** | `pd.to_datetime(..., errors='coerce')` | Enable proper temporal analysis |

| **Missing value imputation** | • **Item** → “Unknown”
• Numeric columns → median
• **Discount Applied** → **False** | Preserve distribution without bias |

| **Validation** | Cleaned rows = 12,575 (unchanged) | Confirms no rows were lost |

4. User Query & Analytical Approach

- **User query:** “Show overall percentage distribution across segments.”
- **Interpretation:** Compute each **Category’s** share of the **total spend** across the entire dataset.
- **Methodology:**

1. **Group** data by **Category** and **sum Total Spent**.
2. **Calculate** overall spend (sum of all categories).
3. **Derive** percentage = (category spend / overall spend) × 100.
4. **Sort** descending for readability.

All operations are performed with Pandas, producing a tidy DataFrame named **result**.

5. Python Analysis Code & Executed Data Output

```
# ----- # 1■ Group by Category & sum Total Spent
category_totals = df.groupby('Category')['Total Spent'].sum() # 2■ Compute overall total
spend overall_total = category_totals.sum() # 3■ Calculate percentage contribution per
Category percentage = (category_totals / overall_total * 100).reset_index(name='Percentage') #
4■ Sort descending for readability percentage = percentage.sort_values('Percentage',
ascending=False).reset_index(drop=True) # 5■ Store final output result = percentage #
-----
```

Executed Output (**result**)

Category	Percentage
Butchers	13.34874707
Electric household essentials	13.10276613
Beverages	12.65766364
Food	12.58653998
Furniture	12.51671453
Computers and electric accessories	12.33187336
Patisserie	11.84498070
Milk Products	11.61071461

(Values are shown to 2■3 decimal places for brevity.)

6. Key Insights & Strategic Recommendations

Insights

1. **Balanced Portfolio:** No single segment dominates; the top three together account for ≈ **39 %** of spend, indicating a fairly diversified revenue base.

2. **Highest■Performing Segment:** *Butchers* leads with **13.35 %**, suggesting strong customer demand or higher price points.
3. **Close Competition:** The gap between the top segment (*Butchers*) and the lowest (*Milk Products*) is only ~ **1.7 %**, highlighting opportunities for cross■segment promotions.

Recommendations

| Action | Expected Benefit |

|-----|-----|

- | **Targeted Marketing for Butchers** – allocate a modest increase in ad spend or loyalty incentives to reinforce the leading position. | Preserve/expand market share in the highest■spending segment. |
- | **Cross■Sell Between Adjacent Segments** – bundle *Beverages* with *Food* or *Electric household essentials* with *Computers*. | Boost average transaction value and smooth demand across categories. |
- | **Pricing Review for Low■Performers** – evaluate price elasticity of *Milk Products* and *Patisserie* to identify margin■improving adjustments. | Potentially raise their contribution without sacrificing volume. |
- | **Seasonality Check** – overlay the distribution with **Transaction Date** to see if certain segments spike at specific times (e.g., holidays). | Enable time■based promotions and inventory planning. |
- | **Continuous Monitoring** – set up a monthly dashboard that recalculates these percentages to detect shifts early. | Proactive response to market changes. |

Prepared by the autonomous AI Data Analyst.

![Show overall percentage distribution across segments. Chart](analysis_chart.png)